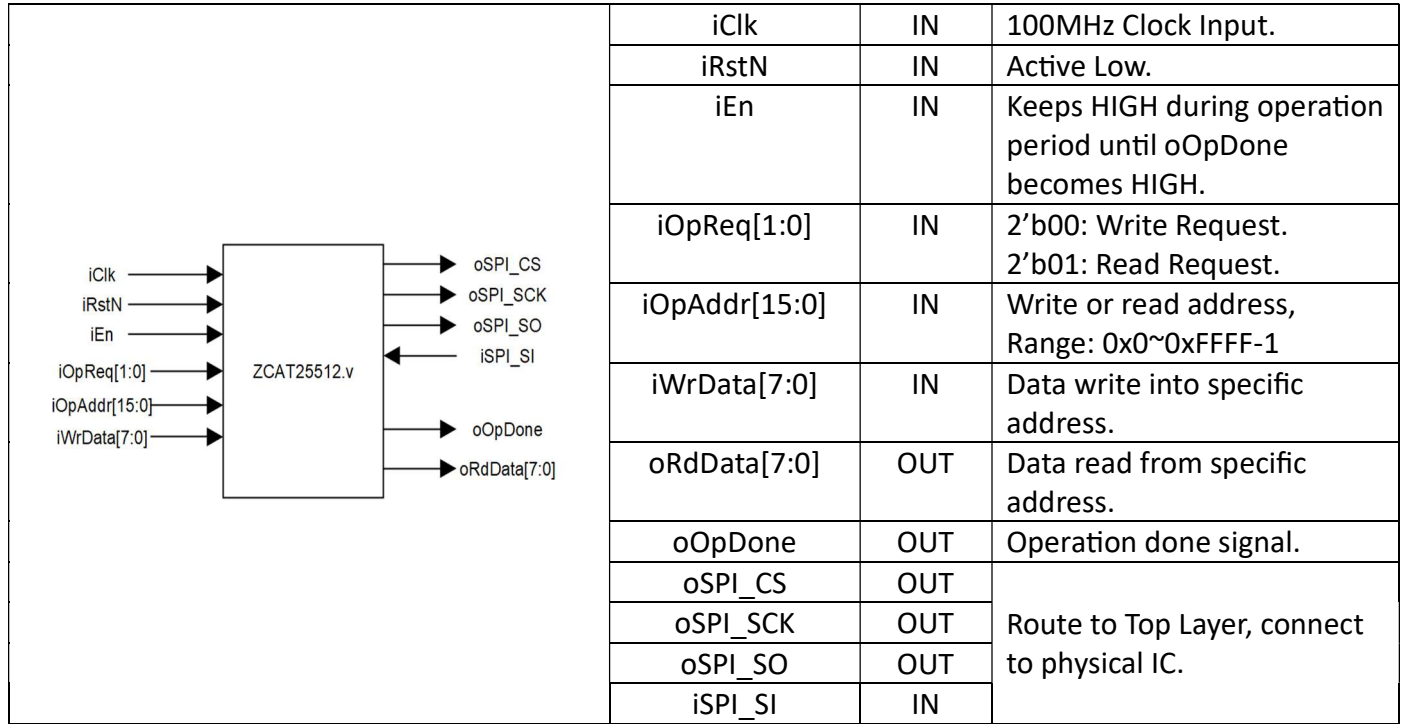


EEPROM CAT5512 IP Manual V0.0.1

1. Block Diagram & Pinout



2. Timing

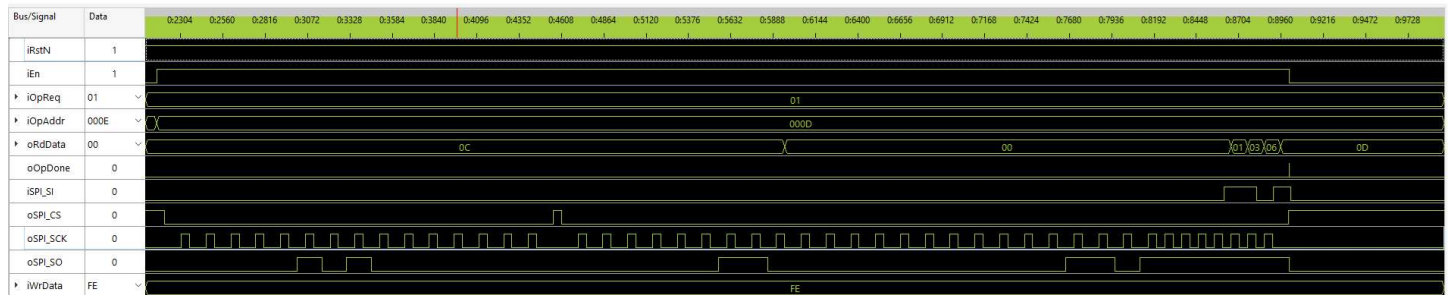
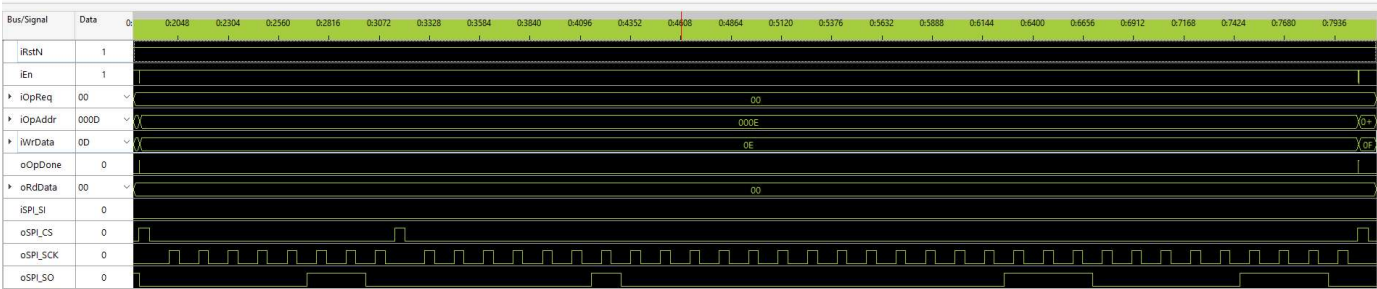
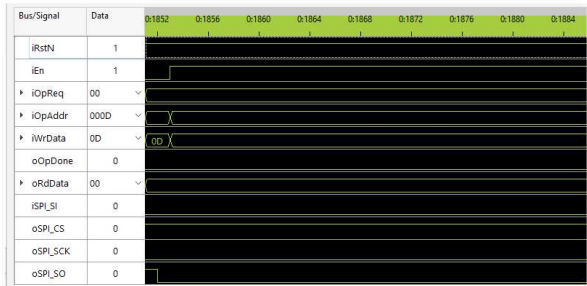


Figure 1. Reading Timing



Details of Start



Details of End

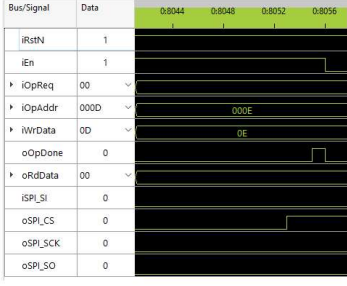


Figure 2: Writing Timing

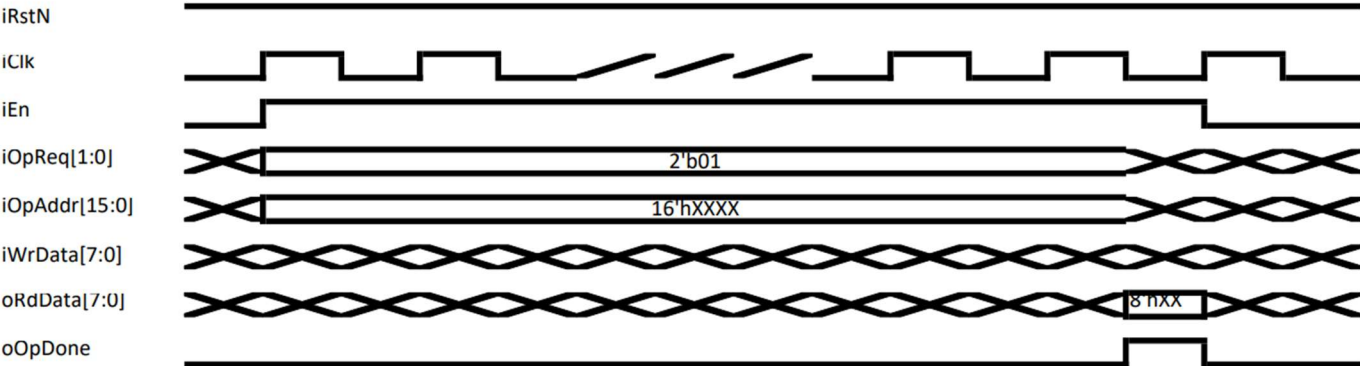


Figure 1. Reading Timing

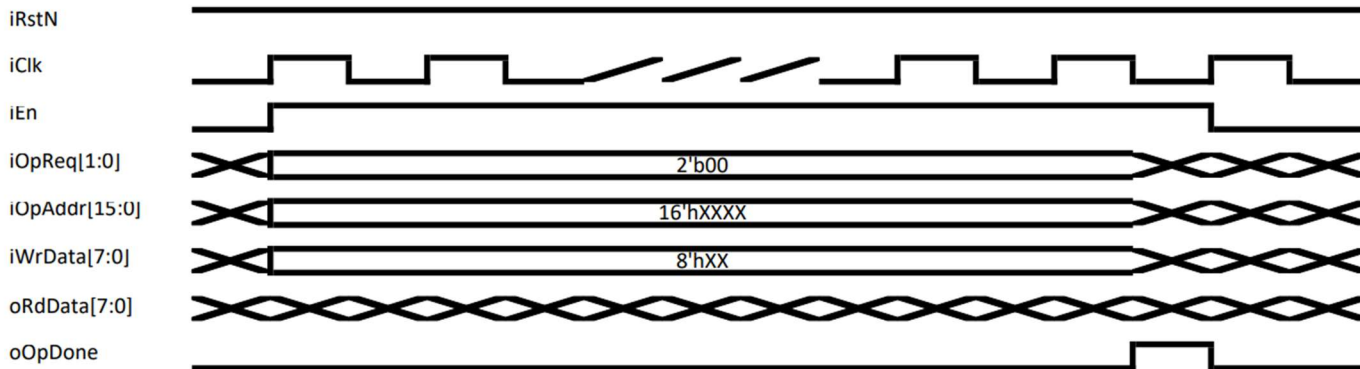


Figure 2. Writing Timing

3. Module Working Principle Brief Description

This module supports SPI compatible EEPROM operating, includes reading and writing. All signals must meet the requirement in Timing diagram. iEn must stays in HIGH during operating period until oOpDone becomes HIGH. Only byte read/write are supported in this version. Burst operation will be provided in next upgrade. Make all input signals ready before pulling up iEn, and keeps them steady during operation. Pull up iEn to start, oOpDone will be pulled up for only single pulse to indicate operation finishes. Once oOpDone is HIGH, pull down iEn immediately.

Since the maximum clock frequency of SPI clock is 5MHz, this module divides the input clock by 50, the ultimate operation frequency is 2MHz.

4. Pin Constraint Contents

```
ldc_set_location -site {L1} [get_ports oSPI_CS]
ldc_set_location -site {M1} [get_ports oSPI_SCK]
ldc_set_location -site {M2} [get_ports oSPI_SO]
ldc_set_location -site {L2} [get_ports iSPI_SI]
ldc_set_port -iobuf {IO_TYPE=LVC MOS18} [get_ports oSPI_CS]
ldc_set_port -iobuf {IO_TYPE=LVC MOS18} [get_ports oSPI_SCK]
ldc_set_port -iobuf {IO_TYPE=LVC MOS18} [get_ports oSPI_SO]
ldc_set_port -iobuf {IO_TYPE=LVC MOS18 PULLMODE=DOWN} [get_ports iSPI_SI]
```

CAUTION!!!

Official specification shows CAT25512 chip only has 1,000,000 Program/Erase Cycles !!!

Continuously Writing the same address will cause cells damaged permanently !!!

Balanced Writing Strategy can be used in top design to extend chips lifecycle.